

Karneth Strata

DOMAIN-AGNOSTIC COORDINATION PLATFORM

TECHNICAL & INDUSTRIAL FEASIBILITY — EXECUTIVE SUMMARY

Karneth Strata is a domain-agnostic coordination platform for autonomous-systems fleets. The same architecture that survives a contested-EMCON kill chain in combat runs a multi-drone wildfire response when comms infrastructure is destroyed. One codebase, mission preset rather than separate product line. This is the executive summary; the companion *Technical Pack v1.1* carries the underlying engineering evidence.

COMPANION DOCUMENT: KS_FEASIBILITY_SEED_TECHNICAL_PACK_V1.1

READ ALONGSIDE: KS INVESTOR BRIEF V1.1

CLASSIFICATION: SHAREABLE EXECUTIVE SUMMARY

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1 Can Karneth Strata actually be built?

The Investor Brief sets out *what* Karneth Strata is and *why now* is the right moment. The natural next question from any serious investor is the feasibility question: **can the company actually build what it describes, on a timeline that matches the funding it is asking for, with the people it has, in the regulatory and supply environment it must operate in?**

The shorter framing for v1.1: KS is not in the business of building better drones. KS is solving the coordination problem that emerges once autonomous systems already exist — in any domain that needs coordinated behaviour under degraded comms. Defence is the scale market. Civil applications (wildfire, disaster response, first responder coordination, infrastructure monitoring) are the same architecture with different mission presets. The feasibility question therefore divides into three classic dimensions and one structural amplifier:

1

TECHNOLOGY FEASIBILITY

The core architecture already runs as a working demonstrator

2

TEAM FEASIBILITY

CTO + COO have shipped at this scale before, in adjacent industries

3

INDUSTRIAL FEASIBILITY

Production ramp anchored by a COO who has done 7-to-250 in twelve months

Each dimension is evidenced in the companion Technical Pack with code, demonstrator screenshots, architectural diagrams, employment records, and risk-register detail. This executive summary is the headline finding from that work — plus the dual-use amplifier described in §3 below, which is new to v1.1.

Karneth Strata is a domain-agnostic coordination platform. Defence missions use EMCON SILENT and ordnance. Wildfire missions use EMCON OPEN and retardant. The architecture in the middle is the same.

The one-line answer: the technical risk is bounded by an already-running demonstrator; the industrial risk is bounded by a COO whose prior career is the exact pattern the company needs to repeat; the procurement risk is mitigated by sovereign stock positioning that the team has been engineering for from the start, not retrofitting; and the single-market risk is reduced by a dual-use operational layer that runs on the same code.

2 The demonstrator is real, and it works

The Karneth Strata command layer exists today as a running simulation platform with hardware-in-the-loop integration. It is not a slide deck; it is a system that an operator can sit down at, command a mixed fleet from, and watch execute autonomous missions in real time. The same code path that runs the simulator drives the live robotic-platform integration used for partner demonstrations.

What is demonstrable today

SINGLE-OPERATOR MULTI-PLATFORM COMMAND

One operator station commanding mixed air and ground unmanned platforms through a unified tactical map, with autonomous mission execution and operator-by-exception intervention. The same operator surface skins to civil mission types via mission-preset configuration.

EMISSIONS-CONTROLLED (EMCON) OPERATION

Full operator displays continue to function under enforced radio silence — every shown value is gated to transmitted-only telemetry, validated end-to-end against an independent audit pass. Defence-grade communications discipline, not a screensaver effect. Civil missions toggle EMCON to OPEN and run the same code with full mesh chatter.

RESILIENT MESH DATALINK

The platform-to-platform communications layer (BURST MESH) sustains command coherence when individual links are degraded or jammed. State is propagated as transmitted serialised diffs, not as shared simulator memory — the architecture is portable to real radios. The same resilience covers post-disaster scenarios where civil cellular infrastructure is destroyed.

HARDWARE-IN-THE-LOOP INTEGRATION

Live robotic integration with PiDog-class ground platforms, including audio direction-of-arrival sensing, distributed kill-chain coordination, and operator-grade after-action review. Demonstrable to visiting partners.

The reviewer can take three things from this. First, the architectural risk is bounded — the hard problems (mixed-platform command, EMCON-coherent operator displays, resilient mesh communications) are not future work, they are demonstrable. Second, the team's engineering discipline shows in the codebase — explicit safety clamps, embedded-grade allocation discipline, audit trails by construction. Third, the path from simulator to fielded system is short because the simulator was built from the start to share code with the real system, not as a separate prototype.

The Technical Pack provides specific evidence for each claim above — code references, demonstrator screenshots, telemetry traces, and the auditing methodology used to verify EMCON discipline.

3 The same architecture, two markets

The KS architecture solves a problem that is identical in shape across defence and civil domains: **coordinated behaviour, under degraded or denied comms, with explicit human-in-the-loop authority, audited end-to-end**. The defence mission is the scale market. The civil mission is the earlier, easier-to-deploy market — and it runs on the same code.

This is not a marketing add-on. The mapping is engineering-literal: the operational concepts of the defence kill chain (detect, corroborate, authorise, engage, assess) map one-to-one onto the operational concepts of a coordinated civil mission. The architecture does not know which domain it is serving. The mission preset does.

Worked example — multi-drone wildfire response

A wildfire in mountainous terrain. Cellular comms have failed because a relay tower is in the fire footprint. The fire incident commander (IC) is at base camp with a KS operator station. Four drones with thermal sensors and retardant payloads are airborne.

PHASE 1	DETECT	Drone A1 sweeps grid pattern; thermal sensor flags a 240 degC hotspot. Pushes contact report onto the platform-shared blackboard.
PHASE 2	CORROBORATE	Drone A2 routes via terrain to confirm. Independent thermal read on the same coordinates. The blackboard now carries two contact reports for the hotspot. Confidence raised; classified as active fire.
PHASE 3	AUTHORISE	The fire IC sees the corroborated hotspot on the tactical map. Approves retardant drop with explicit human-in-the-loop authority. (Civil HITL is, if anything, MORE important than defence HITL because of post-incident liability and audit.)
PHASE 4	STRIKE	Drone A3 is assigned (closest with full retardant load). Mission planner routes through the No-Fly Zones the IC has marked over the evacuated village. Drop executed on coordinates.
PHASE 5	ASSESS	Drone A4 conducts post-drop thermal sweep. Confirms the primary drop hit; flags a residual hotspot at the eastern edge. New contact report loops back to PHASE 2. Cycle continues until BDA reads "fire out".
PHASE 6	RESILIENCE	During the mission, drone A1 ducks behind a ridge and loses line-of-sight to base camp. Drone A4 relays A1's reports via BURST MESH multi-hop. Operator station never loses the picture. The same protocol that handles defence GPS-denied terrain handles civil terrain occlusion.

Every phase above is the same code that runs the defence demonstrator today. The differences are not architectural; they are configuration.

WHAT CHANGES	DEFENCE MISSION	CIVIL (WILDFIRE) MISSION
EMCON preset	SILENT / BURST as doctrine	OPEN (no need for radio silence)
Payload	Ordnance	Fire retardant / water
Operator role	Military commander	Fire incident commander
Mission strings	Combat vocabulary	Response vocabulary
Authority model	Rules-of-engagement chain	Civil command-and-control chain
BDA semantics	Target effect assessment	Fire-out confirmation

The architecture in the middle — the platform-shared blackboard, the BURST MESH datalink, the corroboration logic, the human-in-the-loop authorisation gate, the audit record — is unchanged.

Other civil applications, same architecture

DISASTER RESPONSE

Earthquake, hurricane, flood. The post-event environment is literally comms-denied because infrastructure is destroyed. BURST MESH multi-hop relay was engineered for combat-occluded mesh; it serves post-disaster search-and-rescue coordination identically. The hardest KS engineering is most visible in this use-case.

FIRST-RESPONDER COORDINATION

Multi-agency response (fire, ambulance, police, search teams) needs coordinated drones with shared situational awareness and explicit authority lines. Same coordination platform; mission preset configured for civil emergency rather than combat.

INFRASTRUCTURE MONITORING

Powerlines, pipelines, rail corridors, ports. Recurring-revenue contract model rather than one-shot deployment. The coordination IP serves continuous-monitoring missions as cleanly as it serves incident-response missions.

INDUSTRIAL LOGISTICS & SITE SECURITY

Large industrial sites (mines, ports, energy installations) increasingly procure multi-drone autonomous patrols. The same platform-shared blackboard + HITL authority chain serves this with a different mission preset and payload configuration.

Why this matters for the Series-A reviewer

Civil deployment is likely to happen before defence deployment, for four structural reasons:

1. **Lower regulatory barrier.** UK CAA / EU EASA BVLOS routes for civil unmanned operations are operational today. Defence procurement (UK MoD, DASA, NATO equivalents) is slower and more politically gated.
2. **Faster procurement cycles.** A fire and rescue service or utility operator can procure a coordination platform in months. A defence prime takes years.
3. **Easier field validation.** Civil deployments produce operational telemetry and after-action records that strengthen the defence case when the defence sale comes around. The defence buyer sees a platform that has flown real missions, not just sim runs.
4. **Earlier revenue, earlier datasets.** Civil contracts generate revenue while the defence pipeline matures. This reduces capital burn and de-risks the Series-A from being single-market dependent.

What this means for the reviewer: the dual-use operational layer is not a pivot, a distraction, or a marketing line. It is a property of an architecture that solves a domain-agnostic problem. The investor risk-read is straightforward: the same engineering, the same team, the same capital, and two markets to absorb the work. The civil market gives the platform earlier real-world hours; the defence market gives it scale. Both run on one codebase.

4 The leadership has shipped at this scale before

An autonomous-systems software company at Series-A stage needs three credible answers at the top of the house: **can it engineer the product, can it industrialise it, and can it sell it to defence and regulated buyers?** The Karneth Strata leadership maps to those three answers cleanly.

NIKOLAOS PAPAKOSTAS — CEO

Strategy, corporate structure, investor and customer engagement. Authored the company's strategic capital and positioning work. Detail in the Investor Brief and the SIV / Strategic Capital documents.

THOMAS MAYER — COO

Operations and industrialisation. ETH Zurich (Industrial Management & Manufacturing Engineering) plus MIT Sloan and IMD Lausanne executive programmes. Career detailed in §5.

GREG STARNES — CTO

Engineering and product. Forty years of shipped products in embedded automotive, electric vehicles, autonomous systems, and safety-critical software. Track record detailed in the *Engineering Track Record* document.

The engineering and operations continuity is deep. The CTO, the COO, and the engineering bench (Keith, Joel, Steve) have worked together across multiple companies and many years. That continuity at this depth is unusual at Series-A stage and materially reduces team-formation risk in the first eighteen months of the round. The CEO appointment is recent and deliberate — Nikolaos was introduced through Thomas and brought in to anchor the strategy, capital, and customer-engagement function over the existing engineering-and-operations team. The Investor Brief covers the team in §4.

The engineering bench and the COO have worked with the CTO for years across multiple companies. The CEO is a recent, deliberate appointment on top of that team — strategy, capital, and customer engagement, mapped onto continuity that already existed.

— INTERNAL DILIGENCE FRAMING

5 Why the COO appointment de-risks the ramp

Most defence-software Series-A pitches treat industrialisation as a downstream problem — something to worry about after the technology lands. Karneth Strata is treating it as a present-tense problem because the COO appointment is already in place and the COO's prior career is the exact pattern the company needs to repeat.

Thomas Mayer's career is unusually well-aligned to the KS industrial profile:

BOMBARDIER TRANSPORTATION — TURNAROUND PEDIGREE

As General Manager of Bombardier's Siegen site — Europe's largest bogie manufacturing site, over 500 employees, 2,500-bogie annual output — turned the site from loss-making to industry-best in **one year**. Subsequently Head of Operations across sites in Germany, France, Poland, Hungary, India and China. Implemented lean / Kaizen production system across the entire industrial network.

LOTUS F1 TEAM — HIGH-PRECISION MANUFACTURING UNDER CADENCE

Chief Operating Officer at Lotus F1 / Renault Sport Formula One. Full operative scope: production engineering, procurement, production, logistics, IT, ERP and PLM rollout, board-level budgeting, cash planning with CFO. Carbon-fibre, titanium, aluminium precision manufacturing at race-week cadence — the closest civil-industry analogue to defence-grade unmanned-systems production.

MANOR RACING F1 — PURE SCALE-UP

Chief Executive Officer. Resurrected a Formula One team and scaled it from **seven employees to 250 in twelve months**. First cloud-only F1 operation, with the whole value-add chain run through sub-suppliers. The exact playbook a sovereign-stack drone company needs in years two and three of Series-A capital.

M&A, DUE DILIGENCE, CROSS-BORDER TECH TRANSFER

Multi-million-Euro restructuring programmes, the LEAP global turnaround at Adtranz / Bombardier (~€1B savings over three years), and cross-border technology-transfer projects across Italy, Germany, Poland, India, China, Iran and Russia. Board-level experience at Solaris Bus & Coach.

Earlier career roles include Chief Executive Officer at PeerByte AG / Emobyus, R-Universe (SMTV Holding, Aston Martin dealership and R-ReforGED hypercar manufacturing facility in Warwick), and Ecotive / Bristol Superlight — sustainable-mobility commercial-vehicle ventures.

What this means for the Series-A reviewer:

As a serial entrepreneur, I have demonstrated my ability to build and scale a high-growth, high-margin, high-tech business. I have the proven pattern in-house already. The CEO appointment is not a future hire — it is a present reality. The past two operational years have shown that capital markets and talent markets are not mutually exclusive.

6 The sovereign positioning is engineered, not branded

European defence procurement is moving rapidly away from Chinese-controlled supply chains and US-hosted AI inference. The political pressure is real and irreversible; the supply alternatives are scarce. Karneth Strata is one of the few European defence-software ventures whose architecture is sovereign from the foundation rather than retrofitted around it. The same sovereignty discipline serves civil-domain buyers in regulated sectors (UK CNI, EU member-state critical infrastructure) that are increasingly subject to similar supply-chain-jurisdiction constraints.

That foundation has four parts, each of which is engineered rather than asserted:

1. **Software.** Codebase developed entirely in-house by the KS engineering team. No third-party autonomy-stack dependencies that originate in non-aligned jurisdictions.
2. **AI methodology.** The Lead-Box methodology (Investor Brief §5) keeps models, weights, prompts, source code, mission data, and customer payload local. UK MoD, US ITAR, NATO equivalents, and most primes' subcontract terms forbid customer-touching code from being processed by US-hosted AI APIs. KS can bid on this work; most AI-augmented startups cannot.
3. **Hardware integration.** Platform-agnostic command layer. Customers choose their own airframes, ground platforms, and maritime hulls from aligned suppliers. KS does not lock the buyer to a single hardware vendor.
4. **Industrialisation discipline.** Production ramp planned to Swiss / European industrial standards under a COO whose entire career is European industrial scale-up. Not a contract-assembly play.

The result is a procurement story that is easy for a defence buyer to sign off internally: every layer of the stack is in jurisdictions the buyer can actually inspect, audit, and regulate. This is what European procurement leaders have been asking suppliers for. Few suppliers have it. Civil-domain critical-infrastructure buyers (utility operators, transport operators, port operators) are increasingly held to the same standard.

7 What this seed unlocks in the reviewer's process

This executive summary is the front tier of a two-tier feasibility pack. It is designed to be readable in under fifteen minutes by a non-technical decision-maker and to give that reader enough confidence in feasibility to commit time to the deeper technical evidence.

The companion document — *KS Feasibility Seed Technical Pack v1.1* — carries the engineering evidence behind each headline above: architectural diagrams, demonstrator state, sim-to-real validation methodology, the wildfire-mission worked example in engineering depth, regulatory and safety story, technical roadmap, and a technical-only risk register. It is designed for review by an investor's technical adviser or in-house engineering reviewer.

READER: INVESTOR PRINCIPAL

This document is enough. Use it to decide whether the technical, industrial, and dual-use feasibility is convincing enough to commit a deeper review.

READER: TECHNICAL ADVISER

Proceed to the Technical Pack. The headline claims above are evidenced there with code references, telemetry traces, architecture diagrams, and a worked wildfire mission walkthrough.

READER: DUE-DILIGENCE TEAM

Read both documents alongside the Investor Brief v1.1. The full specialist corpus is then available under extended NDA — see Investor Brief §9.

The feasibility conclusion in one line: the engineering exists, the COO has run this scale-up pattern before, the sovereign positioning is structural, and the architecture is domain-agnostic. The Series-A is buying execution and market access, not invention.

Document: KS_Feasibility_Seed_v1.1.html — Executive Summary (Front Tier)

Companion: KS_Feasibility_Seed_Technical_Pack_v1.1.html — Engineering Evidence (Back Tier)

Read alongside: KS_Investor_Brief_v1.1.html

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